



Ratul Halder

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ABOUT ME

I am a Computer Science graduate with a strong interest in understanding how complex systems work and how data and technology can be used to build reliable, intelligent solutions. I am passionate about solving complex problems using data-driven approaches and building secure, reliable systems. This strongly motivates me to continue my academic development in an environment that values innovation, research and continuous learning.

EDUCATION AND TRAINING

B.Sc in Computer Science and Engineering

Gopalganj Science and Technology University [01/01/2020 – 30/06/2025]

City: Gopalganj | **Country:** Bangladesh | **Website:** <https://gstu.edu.bd/s/> | **Final grade:** 3.60 out of 4.00 | **Level in EQF:** EQF level 6 | **Thesis:** Statistical Reconstruction of Unknown Share in (2,2) Visual Cryptography System

Major Courses: Data Structures, Algorithm Design, Operating Systems, Software Engineering, Computer Architecture, Computer Networks, Calculus, Linear Algebra, Applied Statistics & Queuing, Machine Learning, Artificial Intelligence.

Higher Secondary Certificate

Jessore Govt Shikkha Board Model School & College [01/07/2017 – 06/2019]

City: Jashore | **Country:** Bangladesh | **Field(s) of study:** Science | **Final grade:** 5.00 out of 5.00 | **Level in EQF:** EQF level 4

SKILLS

Programming Languages: Python, HTML, CSS, Flutter / Libraries & Frameworks: Django, Tailwind CSS, Bootstrap, Material UI / Databases: PostgreSQL, MongoDB, Firebase

PROJECTS

Thesis: Statistical Reconstruction of Unknown Share in (2,2) Visual Cryptography System

- Developed a statistical framework to reconstruct unknown shares in (2,2) visual cryptography by exploiting hidden pixel patterns and spatial correlations.
- Engineered an algorithm for candidate share generation using vertical and horizontal distribution analysis to recover structural image features.
- Implemented an iterative refinement process and neighborhood matching to correct local inconsistencies and align shares with known statistical trends.
- Designed a post-processing pipeline utilizing blur filtering and connected component analysis to enhance the visual clarity of reconstructed images.

Link: https://docs.google.com/document/d/1wd44SDX9cV_2jJlnkGsmHBhPLsLHkylAWvfppxgGDhw/edit?usp=sharing

Task Management System

- Secured user authentication and authorization, ensuring that only authorized users can access and modify task data for enhanced security.
- Task Tracking System that simplifies the assignment, updating and management of tasks improving team efficiency and collaboration.
- Implemented role-based task management, granting admins exclusive access to assign and update tasks, set priorities and define deadlines while providing team members with necessary task details.
- Integrated real-time email notifications, ensuring users receive timely alerts for task assignments, status changes and updates facilitating better communication.
- Enabled task progress tracking with real-time status updates (Pending, In Progress, Done) allowing users to monitor the progress and identify bottlenecks.

Technologies: Django, Python, PostgreSQL, HTML, CSS (Tailwind, Bootstrap), Django ORM, Email Integration

Links: <https://ray008ratul.pythonanywhere.com/> | <https://github.com/r258369/TaskTracking/tree/master>

AI Learning Platform

- Using Firebase Auth, ensuring only authorized access to personalized learning data.
- Personalized onboarding quiz collects learning style, subjects, skill level and goals to generate a custom 2-week syllabus with Gemini AI.
- Automated study notes and assignments generated per topic using Gemini AI converted to optimized PDFs, and stored in Firestore subcollections for scalable access.
- Intelligent YouTube video recommendations using Gemini AI for each topic prioritized by subject and channel, with progress tracking for completed videos.
- Dynamic quiz generation and evaluation using Gemini AI, with instant feedback, performance tracking, and quiz history.
- Comprehensive course progress tracking and user profile management, including profile image, email, and password updates.

Technologies: Django, Python, Firebase (Firestore, Auth, Storage), Gemini AI, WeasyPrint, AJAX

Links: <https://ailearning-5n60.onrender.com/> | <https://github.com/r258369/AI Learning>

Code Generation Assistant

- Using data cleaning and preprocessing system that removes corrupted code, normalizes whitespace, filters invalid syntax, and ensures dataset quality.
- Custom GPT-2 model trained on Python code with ByteLevelBPE tokenizer for specialized code generation and understanding.
- Fine-tuned transformer architecture with optimized token embeddings and 52K vocabulary for efficient language modeling.
- Advanced tokenization pipeline ensuring robust handling of programming syntax and edge cases.
- Scalable dataset and training pipeline with automatic tokenization, batching, and checkpoint-based resumable training.

Link: <https://github.com/r258369/PyGPT.git>

VOLUNTEERING

[01/01/2023 – 02/2025] Gopalgonj

Executive Member of GSTU Robotics Club

- Assisted in programming microcontrollers and integrating sensors and hardware components.
- Supported the organization of events, competitions, and training sessions for new members.
- Enhanced problem-solving, teamwork, and project-based learning skills through practical robotics projects.

RECOMMENDATIONS

Name: **Abu-Bakar Muhammad Abdullah** | Assistant Professor
Thesis Supervisor
Email: abdullahcse52@gstu.edu.bd | Phone number: (+880) 1556321671

Name: **Dr. Mrinal Kanti Baowaly** | Chairman of the Department
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